## 21.10. TANK STORAGE OF HIGH LEVEL LIQUID WASTES

The main part of the HLLW is aqueous raffinate from the PUREX cycle. It contains ~99.9% of the nonvolatile FPs, <0.5% of the uranium, <0.2% of the plutonium, and some corrosion products. For each ton of uranium reprocessed about 5 $m^3$ of HLLW is produced. This is usually concentrated to 0.5–1 $m^3$ for interim tank storage; specific activity is in the range $10^7$ GBq $m^{-3}$. The amounts of various elements in the waste and their concentration in 0.5 $m^3$ solution is shown in Table 21.9. The $HNO_3$ concentration may vary within a factor of 2 depending on the concentration procedure. The metal salt concentration is ~0.5 M; it is not possible to keep the salt in solution except at high acidity. The amounts of corrosion products, phosphate, and gadolinium (or other neutron poison added) also may vary considerably. Wastes from the HTGR and FBR cycles are expected to be rather similar.

The HLLW stainless steel tanks have a volume of 50–500 $m^3$. They are rather elaborate (Fig. 21.18); they contain arrangements for cooling (≤65°C to reduce corrosion) and stirring, removal of radiolytic gases, and for control of liquid level, pH, and radioactivity. The tanks are usually double-walled, have heavy concrete shielding, and are often placed underground. Storage in stainless steel tanks has been used in the last 40 y without failure. The philosophy is that tank storage is only an interim procedure and

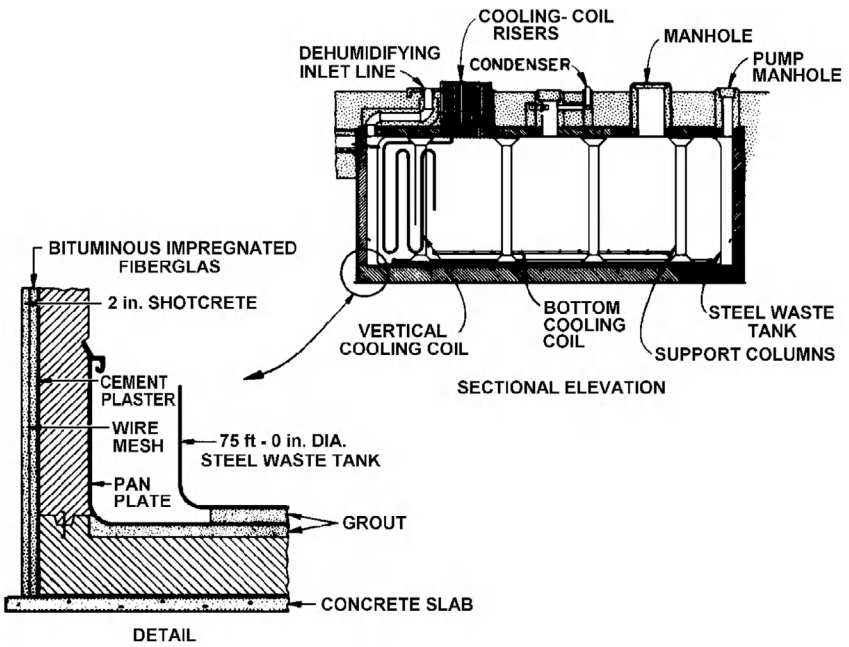



**Figure 21.18** Stainless steel tank for storage of HLLW.

will not last for more than a few years, but the capacity for solidification has often been insufficient, so in practice tank storage has been of much longer duration.

Tank storage has not been without failure. The mild steel tanks built in the 1940's at Hanford, USA, were later (in the 1960's) found to leak due to corrosion. The waste nuclides seeped into the soil, some ultimately reaching ground water; $NO_3^-$, Ru and Cs were found to have moved hundreds of meters, while Sr, Rare Earths and Pu were less mobile in decreasing order.

In 1957 (possibly also in 1967) a serious accident occurred near Chelyabinsk (*Kyshtym*), south of Ekaterinburg, in the former USSR, probably a chemical explosion between organic wastes and nitric acid, in a high-level waste storage facility (tank or underground repository), leading to the contamination of approx. 1600 $km^2$ by $8 \times 10^{16}$ Bq fission products. Local values as high as $10^{10}$ $Bq/m^2$ and values of $2 \times 10^8$ $Bq/m^2$ for $^{90}Sr$ and $^{137}Cs$ have been reported. A large area is now excluded to the public.

## 21.11. OPTIONS FOR FINAL TREATMENT OF HIGH LEVEL WASTES

Many concepts are being studied to treat the high level liquid waste from reprocessing so that the environment is protected against short and long term radiation damage. The final treatment concepts are:

**(a)** dispersion to achieve environmentally acceptable concentrations;
**(b)** partitioning followed by
  **(i)** nuclear transmutation;
  **(ii)** disposal into space;
  **(iii)** burial in a nonaccessible place;
**(c)** solidification followed by geologic deposition.

In principle, dispersion is only applicable for the gaseous and liquid wastes, which would need negligible pretreatment. The limitations are practical (how efficient is the dispersion into air and sea?), radiological (what radioactive concentrations are acceptable in air and sea?) and political/legal (can it be permitted?).

The options (b) are not feasible for all high level waste products, and therefore has to be limited to the most hazardous ones. It would require an isolation of these products, usually referred to as partitioning, or fractionation, of the HLW.

Option (c) is the main route considered for high level wastes from reprocessing. Because of its importance it is discussed in greater detail in a separate subsection (§21.12). Geologic deposition is also the main choice for un-reprocessed spent fuel elements.

### 21.11.1. Dispersion into sea and air

The main danger in release of radioactive waste is the risk of ingestion or inhalation. As the hazard differs considerably between the various waste products, depending on their